

ComLab V3 Emergency Egress Simulation

Concise study guide: how the application was built, how the model works, and what the validation results show.

1. What The Application Does

This application is a Python-powered, browser-based agent simulation for emergency evacuation inside ComLab V3. It compares a current room layout against a modified layout using the same population, fire conditions, and behavior rules.

- Models 41 agents: 36 students, 1 professor, 2 student assistants, and 2 custodians.
- Simulates pack-up delay, locker retrieval, peer waiting, pathfinding, smoke slowdown, trips/faints, door jams, and exit throughput.
- Lets the presenter switch layout mode, panic behavior, fire source, heatmap, speed, and side-by-side comparison.
- Reports total time, evacuated count, average wait, queue length, throughput, exit utilization, trips, door hits, and max heat.

2. How It Was Made

The system is split into a deterministic simulation engine, a local HTTP API, and a canvas dashboard. The same structure can also deploy to Vercel using the public frontend and serverless API.

Layer	Main files	Purpose
Simulation engine	comlab_v3/engine.py	Defines grid constants, agents, fire sources, pathfinding, movement, incidents, and metrics.
Local server/API	comlab_v3/web.py, run.py	Serves the dashboard and exposes /api/state, /api/reset, /api/control, and /api/compare.
Frontend	comlab_v3/static/index.html, app.css, app.js	Draws the room on canvas, handles controls, charts, events, heatmap, and comparison table.
Deployment copy	public/, api/index.py, vercel.json	Static frontend plus Vercel-compatible Python API.
Validation	tests/test_engine.py, scripts/validate_benchmark.py	Regression tests and scenario benchmark matrix.

Implementation Steps

- Created a grid representation of the lab, hallway, doors, workstations, service bay, shelves, TV, and fire extinguishers.
- Defined agent roles and behavior types, then assigned seeded deterministic delays and movement characteristics.
- Implemented breadth-first pathfinding with blocked cells/edges so agents avoid walls, furniture, fire, and service-bay partitions.
- Added incident logic for smoke spread, fire damage, crowd congestion, trips/faints, door jams, and staff assistance.
- Built a canvas dashboard that visualizes the layout, moving agents, fire/smoke, heatmap, metrics, event log, and comparison table.
- Validated every supported fire source in both layouts under panic and non-panic conditions.

3. Simulation Model

Model element	Implementation
Entities	Students, professor, student assistants, custodians.
Resources	Front exit, entrance/back exit, hallway paths, service-bay passage, lockers/shelves, extinguishers.
Events	Pack-up, locker retrieval, peer waiting, movement, extinguisher retrieval, fire suppression, trip/faint, door jam, exit.
Queues	Center aisle and exit approaches are sampled each step for average queue length.
State variables	Time, position, phase, target, path, wait time, active fire cells, heatmap, door cooldown, completion state.
Fire sources	Data rack, instructor desk, student workstation, television, assistant bay.

Core Assumptions

Input	Assumption
Population	41 fixed agents begin inside the room.
Activation	Agents start moving after behavior-based pack-up delays of 2-12 seconds.
Locker service	3 seconds in modified layout; 10-15 seconds in current layout.
Movement	One adjacent grid cell per movement, slowed by smoke, crowding, narrow row gaps, and panic.
Door jams	Unmanaged pressure can block a door for 3-5 steps.
Fire spread	Fire expands through fuel-rich lab cells depending on time, density, heat, and suppression.
Repeatability	Seeded random values keep scenario comparisons deterministic.

Current vs Modified Layout

- Current layout keeps service/storage activity near the entrance path, creating cross-traffic and congestion.
- Modified layout reorganizes the service/storage band, preserves clearer movement lanes, and gives shelf-bound students a defined entrance-door route after retrieval.
- Visual clutter was reduced by removing obvious furniture labels while keeping important safety labels such as EXIT, FIRE EXIT, ENTRANCE, and FE.
- The locker/shelves fire source was replaced by Television as the presentation fire source.

4. How To Demonstrate

Action	What to explain
Start	Shows the live evacuation process and event log progression.
Step	Advances one simulation second for detailed explanation.
Mode	Switches between Current Layout and Modified Layout.
Panic	Changes hesitation, trips, crowding, and door-pressure behavior.
Fire	Moves incident source among data rack, instructor desk, workstation, television, and assistant bay.
Heatmap	Shows accumulated congestion intensity.
Run Comparison	Runs current and modified layouts under the same panic/fire settings and displays metric deltas.

5. Validation And Test Results

The final audit verified Python syntax, deterministic unit tests, API reset behavior, browser control wiring, and the full scenario benchmark matrix.

Check	Result
Unit tests	python -m unittest discover -s tests -> 26 tests OK
Compile check	python -m compileall -q comlab_v3 api tests scripts -> OK
Benchmark matrix	20 scenarios validated: 2 layouts x 2 panic states x 5 fire sources
Evacuation	All 20 benchmark scenarios evacuated 41 / 41 agents before the 400-step cutoff.
Browser/API smoke	Television dropdown, Step button, API reset, and comparison table verified.

Benchmark Summary

Metric	Latest recorded result
Scenarios	20
Total runtime	16.763902 s
Scenarios/sec	1.19
Simulation steps/sec	182.24

Selected Scenario Results

Layout	Panic	Fire	Time	Evac.	Avg wait	Queue	Trips	Doors	Heat
Current	Yes	Data rack	180s	41/41	92.59s	14.43	29	1	778
Modified	Yes	Data rack	71s	41/41	35.59s	18.55	7	2	305
Current	Yes	Instructor	225s	41/41	128.85s	17.95	29	1	918
Modified	Yes	Instructor	86s	41/41	35.05s	15.03	3	0	404
Current	Yes	TV	234s	41/41	122.00s	17.08	22	3	762
Modified	Yes	TV	195s	41/41	108.22s	21.41	5	0	1448
Current	No	Data rack	121s	41/41	60.61s	15.20	13	1	356
Modified	No	Data rack	56s	41/41	24.05s	16.18	2	2	192
Current	No	TV	157s	41/41	92.90s	19.21	12	2	512
Modified	No	TV	262s	41/41	144.32s	20.35	5	0	2112

Interpretation: the modified layout is strongest for the core comparison scenarios, especially data rack and instructor desk fires. The TV scenario is intentionally harder because the fire starts near the front wall route, so it is useful for discussing model limitations and emergency-route sensitivity.

6. Talking Points For Presentation

- This is an agent-based micro-simulation, not a real evacuation certification tool.
- The model is deterministic for a fixed scenario, so results are repeatable during presentation.
- The modified layout generally reduces total time, trips, and door incidents in the main comparison scenarios.
- The dashboard demonstrates cause and effect: fire source, panic, layout, and heatmap all change congestion patterns.
- The strongest recommendation is to keep storage/service activity away from exit approaches and maintain clear guided paths.

7. Quick File Map

File	Study note
run.py	Starts the local simulation web app.
comlab_v3/engine.py	Best file to explain the model, constants, agents, pathfinding, fire, and metrics.
comlab_v3/web.py	Local HTTP API and state service used by the dashboard.
comlab_v3/static/app.js	Canvas drawing, UI controls, chart, events, heatmap, and comparison rendering.
tests/test_engine.py	Regression tests proving reachability, layout rules, evacuation completion, and scenario behavior.
scripts/validate_benchmark.py	Scenario matrix used to produce validation and benchmark results.
public/ and api/index.py	Vercel deployment frontend and serverless API.

One-sentence summary:

The application was built by combining a deterministic Python grid-based agent simulation with a browser canvas dashboard, then validating all supported fire scenarios to compare current and modified ComLab V3 evacuation performance.